

**SAFETY DATA SHEET**

Creation Date 04-November-2019

Revision Date 05-August-2022

Revision Number 4

**1. Identification**

**Product Name** Sorafenib tosylate

**Cat No. :** AC465710000, AC465710010, AC465712500

**CAS-No** 475207-59-1  
**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

**Details of the supplier of the safety data sheet****Company**

**Importer/Distributor**  
Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Manufacturer**  
Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

**Emergency Telephone Number**For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887**2. Hazard(s) identification****Classification****WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                             |                             |
|-------------------------------------------------------------|-----------------------------|
| <b>Reproductive Toxicity</b>                                | Category 1B                 |
| <b>Effects on or via lactation</b>                          | Effects on or via lactation |
| <b>Specific target organ toxicity - (repeated exposure)</b> | Category 1                  |

**Label Elements****Signal Word**

Danger

**Hazard Statements**

May damage fertility or the unborn child

Causes damage to organs through prolonged or repeated exposure

**Precautionary Statements****Prevention**

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Do not breathe dust/fumes/gas/mist/vapours/spray  
Avoid contact during pregnancy/while nursing  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF exposed or concerned: Get medical advice/attention

**Storage**

Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Very toxic to aquatic life with long lasting effects

### 3. Composition/Information on Ingredients

| Component                                                                                                                                            | CAS-No      | Weight % |
|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------|
| 2-Pyridinecarboxamide,<br>4-[4-[[[4-chloro-3-(trifluoromethyl)phenyl]amino]-car<br>bonyl]amino]phenoxy]-N-methyl-,4-methylbenzenes<br>ulfonate (1:1) | 475207-59-1 | <= 100   |

### 4. First-aid measures

|                                        |                                                                                                                   |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                  | If symptoms persist, call a physician.                                                                            |
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.   |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician. |
| <b>Inhalation</b>                      | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.      |
| <b>Ingestion</b>                       | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.             |
| <b>Most important symptoms/effects</b> | None reasonably foreseeable.                                                                                      |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                             |

### 5. Fire-fighting measures

|                                       |                          |
|---------------------------------------|--------------------------|
| <b>Unsuitable Extinguishing Media</b> | No information available |
|---------------------------------------|--------------------------|

|                                         |                          |
|-----------------------------------------|--------------------------|
| <b>Flash Point</b>                      | No information available |
| <b>Method -</b>                         | No information available |
| <b>Autoignition Temperature</b>         | No information available |
| <b>Explosion Limits</b>                 |                          |
| <b>Upper</b>                            | No data available        |
| <b>Lower</b>                            | No data available        |
| <b>Sensitivity to Mechanical Impact</b> | No information available |
| <b>Sensitivity to Static Discharge</b>  | No information available |

**Specific Hazards Arising from the Chemical**

Do not allow run-off from fire-fighting to enter drains or water courses.

**Hazardous Combustion Products**

Carbon oxides. Nitrogen oxides (NOx). Hydrogen fluoride. Hydrogen chloride gas.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
2

**Flammability**  
1

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                                             |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.                                                                                                                                           |
| <b>Environmental Precautions</b>            | Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.                                                                                                                                |

## 7. Handling and storage

|                 |                                                                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation. |
| <b>Storage.</b> | Keep container tightly closed in a dry and well-ventilated place. To maintain product quality. Store in freezer. Incompatible Materials. Oxidizing agent.                           |

## 8. Exposure controls / personal protection

|                            |                                                                                                                                              |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Guidelines</b> | This product does not contain any hazardous materials with occupational exposure limitsestablished by the region specific regulatory bodies. |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|

**Engineering Measures**

Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment**

|                        |                                                                           |
|------------------------|---------------------------------------------------------------------------|
| <b>Eye Protection</b>  | Goggles                                                                   |
| <b>Hand Protection</b> | Wear appropriate protective gloves and clothing to prevent skin exposure. |

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Nitrile rubber | See manufacturers | -               | Splash protection only |
| Neoprene       | recommendations   |                 |                        |
| Natural rubber |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

#### Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                        |                                                                                                                                  |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Physical State                         | Solid                                                                                                                            |
| Appearance                             | White to brown                                                                                                                   |
| Odor                                   | No information available                                                                                                         |
| Odor Threshold                         | No information available                                                                                                         |
| pH                                     | No information available                                                                                                         |
| Melting Point/Range                    | No data available                                                                                                                |
| Boiling Point/Range                    | No information available                                                                                                         |
| Flash Point                            | No information available                                                                                                         |
| Evaporation Rate                       | Not applicable                                                                                                                   |
| Flammability (solid,gas)               | No information available                                                                                                         |
| Flammability or explosive limits       |                                                                                                                                  |
| Upper                                  | No data available                                                                                                                |
| Lower                                  | No data available                                                                                                                |
| Vapor Pressure                         | No information available                                                                                                         |
| Vapor Density                          | Not applicable                                                                                                                   |
| Specific Gravity                       | No information available                                                                                                         |
| Solubility                             | Insoluble in water                                                                                                               |
| Partition coefficient; n-octanol/water | No data available                                                                                                                |
| Autoignition Temperature               | No information available                                                                                                         |
| Decomposition Temperature              | No information available                                                                                                         |
| Viscosity                              | Not applicable                                                                                                                   |
| Molecular Formula                      | C <sub>21</sub> H <sub>16</sub> Cl F <sub>3</sub> N <sub>4</sub> O <sub>3</sub> . C <sub>7</sub> H <sub>8</sub> O <sub>3</sub> S |
| Molecular Weight                       | 637.03                                                                                                                           |

## 10. Stability and reactivity

|                 |                                              |
|-----------------|----------------------------------------------|
| Reactive Hazard | None known, based on information available   |
| Stability       | Stable under recommended storage conditions. |

|                                         |                                                                                |
|-----------------------------------------|--------------------------------------------------------------------------------|
| <b>Conditions to Avoid</b>              | Incompatible products.                                                         |
| <b>Incompatible Materials</b>           | Oxidizing agent                                                                |
| <b>Hazardous Decomposition Products</b> | Carbon oxides, Nitrogen oxides (NOx), Hydrogen fluoride, Hydrogen chloride gas |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                       |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                  |

## 11. Toxicological information

### Acute Toxicity

**Product Information** Caution - substance not yet tested completely

### Component Information

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** Irritating to eyes and skin May cause irritation of respiratory tract

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                                                                                         | CAS-No      | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|---------------------------------------------------------------------------------------------------|-------------|------------|------------|------------|------------|------------|
| 2-Pyridinecarboxamide                                                                             | 475207-59-1 | Not listed | Not listed | Not listed | Not listed | Not listed |
| 4-[4-[[[4-chloro-3-(trifluoromethyl)phenyl]amino]phenoxy]-N-methyl-4-methylbenzenesulfonate (1:1) |             |            |            |            |            |            |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

**Persistence and Degradability** Insoluble in water

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment due its low water solubility.

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

**UN-No** UN3077  
**Proper Shipping Name** ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S.  
**Technical Name** 2-Pyridinecarboxamide,  
 4-[4-[[[4-chloro-3-(trifluoromethyl)phenyl]amino]-carbonyl]amino]phenoxy]-N-methyl-,4-methylbenzenesulfonate (1:1)  
**Hazard Class** 9  
**Packing Group** III

#### TDG

**UN-No** UN3077  
**Proper Shipping Name** ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S.  
**Hazard Class** 9  
**Packing Group** III

#### IATA

**UN-No** UN3077  
**Proper Shipping Name** ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S.  
**Hazard Class** 9  
**Packing Group** III

#### IMDG/IMO

**UN-No** UN3077  
**Proper Shipping Name** ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S.  
**Hazard Class** 9  
**Packing Group** III

### 15. Regulatory information

**All of the components in the product are on the following Inventory lists:** X = listed U.S.A. (TSCA) Canada (DSL/NDSL) Europe (EINECS/ELINCS/NLP) Australia (AICS) Korea (KECL) China (IECSC) Japan (ENCS) Philippines (PICCS) Taiwan (TCSI) Japan (ISHL) New Zealand (NZIoC) Japan (ISHL)

#### International Inventories

| Component                                                                                                                                  | CAS-No      | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS | ELINCS | NLP |
|--------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----|------|------|-----------------------------------------------|--------|--------|-----|
| 2-Pyridinecarboxamide, 4-[4-[[[4-chloro-3-(trifluoromethyl)phenyl]amino]-carbonyl]amino]phenoxy]-N-methyl-,4-methylbenzene sulfonate (1:1) | 475207-59-1 | -   | -    | -    | -                                             | -      | -      | -   |

| Component                                                                                                                                  | CAS-No      | IECSC | KECL | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|--------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------|------|------|------|------|------|-------|-------|
| 2-Pyridinecarboxamide, 4-[4-[[[4-chloro-3-(trifluoromethyl)phenyl]amino]-carbonyl]amino]phenoxy]-N-methyl-,4-methylbenzene sulfonate (1:1) | 475207-59-1 | -     | -    | -    | X    | X    | -    | -     | -     |

#### Legend:

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)  
**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory  
**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances  
**IECSC** - Chinese Inventory of Existing Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances  
**ENCS** - Japanese Existing and New Chemical Substances  
**AICS** - Australian Inventory of Chemical Substances  
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

## Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

## Other International Regulations

### Authorisation/Restrictions according to EU REACH

### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                                                                                                                                  | CAS-No      | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|--------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| 2-Pyridinecarboxamide, 4-[4-[[[4-chloro-3-(trifluoromethyl)phenyl]amino]-carbonyl]amino]phenoxy]-N-methyl-, 4-methylbenzenesulfonate (1:1) | 475207-59-1 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                                                                                                                                  | CAS-No      | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|--------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| 2-Pyridinecarboxamide, 4-[4-[[[4-chloro-3-(trifluoromethyl)phenyl]amino]-carbonyl]amino]phenoxy]-N-methyl-, 4-methylbenzenesulfonate (1:1) | 475207-59-1 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

**Prepared By** Regulatory Affairs  
 Thermo Fisher Scientific  
 Email: EMSDS.RA@thermofisher.com

**Creation Date** 04-November-2019  
**Revision Date** 05-August-2022  
**Print Date** 05-August-2022  
**Revision Summary** Not applicable.

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**